

Medical Laboratory Science / Medical Laboratory Technology

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Question 6

Not yet answered

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 [Flag question](#)

A 25-year-old patient came to the laboratory with a request of coagulation profile test. The laboratory technologist arranged necessary materials for blood collection. The technician planned to collect the blood with 0.5ml of sodium citrate containing test tube. What volume of blood should proportionally be collected?

- ☐ a. 18ml
- ☒ b. 4.5ml
- ☐ c. 9ml
- ☐ d. 5ml

Medical Laboratory Technology

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Question 5

Not yet answered

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A 25-year-old male patient visited a hospital for his clinical feature of diffuse infiltration of the skin by multiple nodules of varying size. The patient mentioned as he has family history of leprosy. The laboratory diagnosis of the specimen taken from skin nodule showed numerous acid fast rods. What is the possible route of transmission of this pathogen?

- ☐ a. Vector borne
- ☐ b. Contaminated water
- ☒ c. Contact with lesion
- ☐ d. Hereditary

[Clear my choice](#)

Time left 2:57:55

Question 4

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 [Flag question](#)

A 23-year-old female patient with history of chronic urinary tract infection visited a physician for follow-up. The physician physically diagnosed and advised her to give urine specimen. Then, the laboratory technologist received urine sample and inoculated it on culture media. What is the preferable urine specimen to be used for the above test?

- ☐ a. Post-prandial urine specimen
- ☐ b. 24-hour urine specimen
- ☐ c. Random urine specimen
- ☒ d. Mid-stream urine specimen

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Question 3

Answer saved

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A 20-year-old man who has HIV/AIDS, was admitted in an emergency room with complains of meningitis. The physician who observes the patient sent CSF lumbar puncture to a laboratory for microbiological examination. What will be the most possible microscopic method that should be done?

- ☐ a. Aiding 10 percent KOH on CSF and observing with 40x
- ☒ b. Observing agglutination of serum antibody and antigen with 40x
- ☐ c. Observing direct wet mount with 40x
- ☐ d. Mixing CSF with Indian ink and observing with 40x

[Clear my choice](#)

Medical Laboratory Science / Medical Laboratory Technology

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Question 2

Not yet answered

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A 20-year-old boy, who developed a well-demarcated skin decolourization on his chest, visits a clinician. The clinician suspected that it could be due to fungal infection and requested for laboratory confirmation. What is the appropriate specimen to be collected for diagnosis of this patient?

- ☐ a. Skin snip
- ☐ b. Skin biopsy
- ☐ c. Skin excision
- ☒ d. Skin scraping

[Clear my choice](#)

Time left 2:59:35

Question 1

Not yet answered

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 [Flag question](#)

A 15-year-old boy with skin problem came to a hospital, and he was seen by a dermatologist. On physical examination, a dermatologist noticed lightly pigmented areas on his chest, arms and abdomen. A dermatologist sent the boy to a laboratory and the laboratory technologist collected specimen for microscopic examination. What appropriate step should be followed during specimen collection?

- ☐ a. Clean the skin with tincture iodine and air dry
- ☐ b. Rub the skin with dry cotton or tissue paper
- ☐ c. Clean the skin with 70percent alcohol and air dry
- ☐ d. Clean the skin with water and air dry

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Not yet answered

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 [Flag question](#)

A 35-year-old female patient came to a laboratory for haematological investigations. A phlebotomist has received the request paper and planned to collect venous blood from the patient. What should be the most appropriate vein preference to collect the blood?

- ☐ a. Basilic vein of the forearm
- ☐ b. Cephalic vein of the forearm
- ☒ c. Median cubital vein of the forearm
- ☐ d. Jugular vein of the neck

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Question 10

Not yet answered

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A 30-year-old man, who had a history of irritation and urethral discharge, was requested the specimen to laboratory for gram stain examination. A laboratory worker had performed smear, staining and examination for collected urethral discharge. Finally, the result was reported as "many gram negative intracellular cocci seen/HPF." What bacterial information was missed in the above test result reporting?

- ☐ a. Shape
- ☒ b. Arrangement
- ☐ c. Location
- ☐ d. Load

[Clear my choice](#)

Medical Laboratory Technology

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Question 9

Not yet answered

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 [Flag question](#)

A 29-year-old sexually active woman came to a hospital with complain of dysuria, pyuria, frequency, and urgency. The physician suspected that the infection is caused by Coagulase negative Staphylococcus. What is the most appropriate sample for diagnosis of the disease?

- ☐ a. First void urine
- ☐ b. 24hours urine
- ☐ c. Morning urine
- ☒ d. Clean-catch midstream urine

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Question 8

Not yet answered

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 [Flag question](#)

A 28-year-old pregnant woman with blood type A Rh negative, who is married to a blood type B Rh positive man, came to a hospital for treatment of abdominal discomfort. This is her second pregnancy and she had no any follow up history during her first pregnancy. Hence, the physician performed physical examination and found that the fetus had reduced movement. What type of anaemia is the patient most likely suffering from?

- ☒ a. Haemolyticaemia
- ☐ b. Megaloblasticaemia
- ☐ c. Iron deficiency anaemia
- ☐ d. Aplastic anaemia

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Question 7

Not yet answered

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 [Flag question](#)

A 27-year-old febrile patient came to a laboratory for blood film examination. A laboratory technologist performed skin puncture and collected blood on microscopic slides. Then, the technician prepares thin smear using two clean microscopic glass slides. What method was used to make the thin blood smear?

- ☐ a. Cover glass method
- ☒ b. Wedge method
- ☐ c. Automated method
- ☐ d. Cytospine method

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Question 16

Not yet answered

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 [Flag question](#)

A 35-year-old-female patient with complaints of high fever and joint pain visited a clinician in a given hospital. During laboratory diagnosis, the patient's serum was found icterus, she was positive for plasmodium falciparum and slightly anemic. Her serum direct bilirubin level was 0.2 mg/dl (with reference range 0-0.5 mg/dl), and serum total bilirubin level was 2 mg/dl (with reference range of 1.0-1.5 mg/dl). What kind of jaundice did the result indicate?

- ☒ a. Pre-hepatic jaundice
- ☐ b. Hepatic jaundice
- ☐ c. Post-hepatic jaundice
- ☐ d. No jaundice

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Question 15

Not yet answered

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A 35-year-old patient came to a hospital with major symptoms of severe headache, stiff neck, and vomiting accompanied by delirium and confusion. The clinician suspected third stage meningococcal infections and he wants to collect specimen for laboratory diagnosis, but he fails to decide how and from where to take the sample. What is the most like site and specimen type to be punctured and collected?

- ☐ a. Throat washing to get throat swab
- ☐ b. Vein puncture to collect blood sample
- ☐ c. Nasal aspiration to get nasal secretion
- ☒ d. Lumbar puncture to collect CSF

[Clear my choice](#)

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Question 14

Not yet answered

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 [Flag question](#)

A 35-year-old male with a history of drinking more than 30gm (equivalent to 3 to 4 drinks) alcohol per day was investigated for LFTs and the results showed slight elevations of AST, ALT and GGT and fatty infiltrates in liver biopsy. What is the most likely hepatic disorder diagnosed based on the above findings?

- ☒ a. Alcoholic induced fatty liver diseases
- ☐ b. Non-alcohol induced fatty liver diseases
- ☐ c. Toxic induced fatty liver diseases
- ☐ d. Viral induced fatty liver diseases

[Clear my choice](#)

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Question 13

Not yet answered

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[Flag question](#)

A 35-year-old male patient with complaints of weakness and poor energy visited a clinician. The clinician suspected the patient for malabsorption. Thus, the clinician requested serum protein profile test. Accordingly, the technologist processed the sample and measured serum protein profile level. Which serum protein profile level results confirm the clinician's diagnosis?

- ☐ a. Albumin high, and globulin normal
- ☒ b. Albumin normal, and globulin low
- ☐ c. Albumin low, and globulin normal
- ☐ d. Albumin normal, and globulin high

[Clear my choice](#)

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Question 12

Not yet answered

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 [Flag question](#)

A 35-year-old male patient came to a hospital with complaint of acute febrile illness. The physician suspected that the patient might be infected by Borella species and requested for microscopic examination. What is the appropriate specimen and stain used for examination of the pathogen?

- ☐ a. CSF specimen and AFB stain
- ☒ b. Blood specimen and Geimsa stain
- ☐ c. Stool specimen and Indian ink stain
- ☐ d. Urine specimen and gram stain

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Question 21

Not yet answered

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 [Flag question](#)

A 40-year-old male patient with history of polydipsia, polyphagia and polyuria gave a 24 hours urine specimen for the urinalysis laboratory. The laboratory technologist measured urine volume and recorded the result as above 2000ml. Which is the next most appropriate chemical parameter to be examined?

- ☒ a. Glucose test
- ☐ b. Protein test
- ☐ c. Nitrite test
- ☐ d. Bilirubin test

[Clear my choice](#)

Not yet answered

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 [Flag question](#)

A 40-year-old male patient brought a request for the determination of serum testosterone hormone level. The laboratory technologist collected blood sample and started processing the measurement of testosterone hormone level by ELISA method. Which statement best describes the test principle of the method?

- ☐ a. Degree of product formation between serum analyte and respective reagent
- ☐ b. Degree of refraction formed after a radiant energy passes through a turbid solution
- ☒ c. Degree of interaction of serum Ag or Ab with reagent Ab or Ag, respectively
- ☐ d. Degree of emitted radiant energy when serum analyte returns inert after excitation

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Question 19

Not yet answered

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A 40- year-old farmer living in the rural parts of Ethiopia, who has been walking bare foot for the last 10 years, developed small warty like lesions on his leg. The attending dermatologist assessed patient's history and suspected saprophytic fungal infection acquired from the environment. What is the most likely route of transmission of the disease?

- ☐ a. Direct contact with infected individual
- ☒ b. Inhalation of spores from environment
- ☐ c. Traumatic inoculation of fungal hyphae
- ☐ d. Drinking contaminated river water

Question 18

Not yet answered

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 [Flag question](#)

A 37-year-old woman came to a hospital with extensive destruction of muscle and fat. Clinically, streptococcal gangrene was diagnosed. A tissue sample from the affected body part was sent for isolation of the bacteria. What are the appropriate culture media used to isolate the suspected bacteria?

- ☐ a. Xylose Lysine Deoxycholate agar and Hektion enteric agar
- ☒ b. Blood agar and chocolate agar
- ☐ c. Mannitol salt agar and MacConkey agar
- ☐ d. Thiosulphate-citrate bile salt sucrose and Cystine lactose electrolyte-deficient agar

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Question 17

Not yet answered

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 [Flag question](#)

A 37-year-old male patient with bradycardia, decreased pulse pressure, cool and/or dry skin, puffy eyes, and loss of the outer lateral eyebrows visited a laboratory for thyroid function tests. Then, the laboratory technologist collected and processed the sample according to the established SOPs. The sample was analyzed using chemiluminescence principle and the following laboratory results were reported; high TSH, low free T4, and low T3. What disease condition most likely caused the above alterations?

- ☒ a. Primary hypothyroidism
- ☐ b. Tertiary hypothyroidism
- ☐ c. Secondary hypothyroidism
- ☐ d. Central hypothyroidism

[Clear my choice](#)



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Flag question

A 4th year medical laboratory student wanted to study the prevalence of malaria among pregnant women for his graduation research. The student prepared 10percent Giemsa stain working solution to stain thin and thick blood films prepared from pregnant women via skin puncture. During examination, the student observed that few of the slides appeared to be too blue under light microscope. What is the most likely cause of the problem in this procedure?

- ☐ a. Too thin films
- ☒ b. Inadequate washing of the stain
- ☐ c. Insufficient staining time
- ☐ d. Too high acidity of the stain

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Time left 2:50:08

Question 25

Not yet answered

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 [Flag question](#)

A 45-year-old man with frequent bleeding in the nose came to a referral hospital. The internist requested PT, INR and APTT to the laboratory. The laboratory technologist collected blood with citrate test tube and ran PT and INR tests. Before performing APTT, the technologist added kaolin suspension to the APTT reagent. What is the effect of kaolin in the APTT test?

- ☒ a. Activator
- ☐ b. Anticoagulant
- ☐ c. Decelerator
- ☐ d. Inhibitor

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Question 24

Not yet answered

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[Flag question](#)

A 45-year-old male patient with chronic glomerulonephritis and long lasting extremities edema came to the laboratory for serum albumin and urinalysis tests. The laboratory technologist received and analyzed the sample and result was serum creatinine of 1.7 mg/dl (RI; 0.7-1.4g/dl) and serum albumin of 15g/l (RI; 30-45g/l). What would happen to the concentration of urinary protein?

- ☒ a. It will be increased
- ☐ b. There will be no protein
- ☐ c. It will be decreased
- ☐ d. It will be normal

[Clear my choice](#)

Medical Laboratory Science / Medical Laboratory Technology

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Question 23

Not yet answered

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[Flag question](#)

A 42-year-old female patient came to a laboratory for renal function tests. The patient ate purine rich diet for her breakfast before arrived at the laboratory. The laboratory technologist collected, processed and analyzed the blood sample. Which parameter was most likely affected?

- ☐ a. Creatinine
- ☐ b. Ammonia
- ☒ c. Uric acid
- ☐ d. Urea

[Clear my choice](#)



A 40-year-old man and his family had presented at a health center with, fever, muscle pains, nausea, vomiting, abdominal cramps, and severe diarrhea and dehydration. The physician admitted all the families in isolated room and took immediate action to manage the dehydration. A rice water stool culture was requested for more investigation of the pathogen and at the same time strict safety precaution was applied in the health facility in order to control spread of the pathogen. What would be the most likely route of transmission of the causative agent?

- ☐ a. Contact with soil contaminated cloths
- ☐ b. Contaminated hand with fecal matter
- ☒ c. Drinking water contaminated with fecal
- ☐ d. Contac with infected animals

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Question 32

Not yet answered

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[Flag question](#)

A blood film examination result of a 52-year-old man indicated trophozoite stage of *P.falciparum*, decreased blood hemoglobin level and reduced total RBC. The laboratory technologist also observed red urine. What is the most likely urine parameter associated with the above findings?

- ☐ a. Bilirubin
- ☐ b. Hemoglobin
- ☐ c. Glucose
- ☐ d. Myoglobin

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Question 31

Not yet answered

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[Flag question](#)

A 72-year-old male patient brought a request paper for lipid profile tests to the laboratory. The laboratory technologist collected blood specimen to perform the above tests. The results of the lipid profile tests showed high total cholesterol, low HDL cholesterol and high LDL cholesterol. What is the most likely underlying disease of the patient?

- ☒ a. Cardiovascular disease
- ☐ b. Pulmonary disease
- ☐ c. Renal disease
- ☐ d. Gastro intestinal disease

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Question 30

Not yet answered

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 [Flag question](#)

A 70-year-old male patient with history of renal complications came to urinalysis laboratory for routine urinalysis tests. The examination results showed that the urine was significantly turbid and positive for urinary protein. What shall the technologist expect from the microscopic examination?

- ☐ a. Urinary crystal
- ☒ b. Urinary casts
- ☐ c. Parasite
- ☐ d. Bacteria

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Question 29

Not yet answered

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A 65-year-old patient, with symptoms of shortness of breathing, chest pain (angina pectoris) and weakness suspected for coronary heart disease (CHD) was requested for lipid profile laboratory investigations. The lipid profile results revealed elevated values of HDL-C, LDL-C, triglyceride and total cholesterol. Which lipid profile is the most risk factor for atherosclerosis?

- ☐ a. Elevated HDL-C
- ☐ b. Elevated total cholesterol
- ☐ c. Elevated LDL-C
- ☐ d. Elevated triglyceride

Medical Laboratory Technology

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Question 28

Not yet answered

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A 64-year-old man develops a small, painless, subcutaneous nodule on his foot that increases slowly but progressively in size. Microscopic examination revealed, dark colored branching intertwining tubes with separation in between and gram-positive partially branching beaded filaments. What is the most likely pathogen detected from the above patients?

- ☐ a. Virus
- ☐ b. Parasite
- ☒ c. Fungus
- ☐ d. Bacteria

[Clear my choice](#)

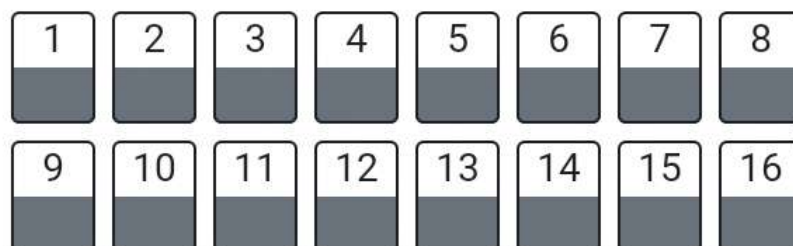
[Flag question](#)

A 60-year-old patient suspected of viral hepatitis came to a referral laboratory. The technologist collects blood specimen and send it to molecular virology laboratory. Genomic analysis and electron microscopy were done. The report showed enveloped, small, partly double stranded DNA virus. What is the most likely hepatitis virus identified?

- ☒ a. Hepatitis B Virus (HBV)
- ☐ b. Hepatitis D Virus (HDV)
- ☐ c. Hepatitis C Virus (HCV)
- ☐ d. Hepatitis A Virus (HAV)

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Question 36

Not yet answered

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 [Flag question](#)

A fourth year medical laboratory student wanted to measure the hemoglobin level of under five children in the community for research purpose using a hemoCue method. The student made skin puncture and a small drop of capillary blood was attracted by capillary action into a specially designed single used microcuvette which contains dry reagents (sodium azide, and sodium nitrite). What is the most likely role of sodium nitrite and sodium azide in this procedure?

- ☒ a. To decrease the turbidity in the solution
- ☐ b. To disintegrate the white blood cells
- ☐ c. To convert hemoglobin to methemoglobin
- ☐ d. To disintegrate the red cell membrane

[Clear my choice](#)

Medical Laboratory Technology

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Question 35

Not yet answered

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[Flag question](#)

A female prisoner developed an acute febrile illness after three weeks of her imprisonment. She visited the prison clinic for help. A physician suspected for epidemic typhus and requested for serological Weil-Felix test and the laboratory technologist conducted the test for confirmation. What type of antigen could be most likely used as a reagent for the test?

- ☐ a. Chlamydia somatic antigen
- ☒ b. Proteus somatic antigen
- ☐ c. Salmonella somatic antigen
- ☐ d. Rickettsia somatic antigen

[Clear my choice](#)



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 [Flag question](#)

A clinician wants to diagnose a neonate born from an HIV-infected mother. The birth attending clinician collects blood sample and send it to a laboratory. The laboratory technologist performed rapid serological (antibody) test and the result was positive. What is the most appropriate next step in the diagnosis of this infant?

- ☐ a. Repeating the test by another serologic test
- ☐ b. Reporting the result as positive
- ☒ c. Collecting another sample for RT-PCR
- ☐ d. Reporting the result as indeterminate

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Time left 2:46:02

Question 34

Not yet answered

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[Flag question](#)

A clinician wants to diagnose a neonate born from an HIV-infected mother. The birth attending clinician collects blood sample and send it to a laboratory. The laboratory technologist performed rapid serological (antibody) test and the result was positive. What is the most appropriate next step in the diagnosis of this infant?

- ☐ a. Repeating the test by another serologic test
- ☐ b. Reporting the result as positive
- ☒ c. Collecting another sample for RT-PCR
- ☐ d. Reporting the result as indeterminate

[Clear my choice](#)

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Question 33

Not yet answered

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 [Flag question](#)

A cardiac biomarker designated as "Y" is the biomarker whose basic test principle is determined by immunoinhibition that allows blocking of one isotope of the "Y" isoenzyme found in patient samples not to react with a blocking reagent while allowing only the other component to react with the reagent forming NADH that is directly proportional to the activity of the isoenzyme of the biomarker present in patient samples and measured spectrophotometrically at 340nm. Which cardiac biomarker is best principled based on the above given information?

- ☐ a. CK-BB
- ☐ b. CK-MB
- ☐ c. LDH1
- ☐ d. CK-MM

Question 40

Not yet answered

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 [Flag question](#)

A laboratory technologist collected blood specimen from a 32-year-old male patient for the diagnosis of prostate cancer. The technologist processed and analyzed the sample for prostate specific antigen using two antibody techniques. The first antibody was immobilized in the micro particle and the second antibody was labeled with enzyme. There was a color change when substrate was changed to product by enzyme and the absorbance of colored compound was measured spectrophotometrically. Which method was used by the technologist?

- ☐ a. Radio labeled immune assay
- ☐ b. Enzyme labeled immunocytochemistry
- ☐ c. Electro chemiluminescence immunoassay
- ☐ d. Enzyme linked immunosorbent assay

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Question 39

Not yet answered

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 [Flag question](#)

A laboratory technologist collected blood specimen from a 22-year-old male patient for AST measurement. The chemistry analyzer was under maintenance for about 10 hours and the technologist centrifuged the blood sample to store the serum sample for 48 hours. What was the most appropriate storage temperature?

- ☐ a. 8 degree centigrade
- ☐ b. 10 degree centigrade
- ☒ c. 4 degree centigrade
- ☐ d. 20 degree centigrade

[Clear my choice](#)



Medical Laboratory Technology

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Question 38

Not yet answered

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[Flag question](#)

A laboratory technician collected blood from the vein of a 25-year-old male patient. While the technician was collecting blood from the patient, a large swelling formed around the puncture site was observed. What is the most likely complication encountered in the above scenario?

- ☐ a. Hemoconcentration
- ☐ b. Hemolysis
- ☒ c. Hematoma
- ☐ d. Ecchymosis

[Clear my choice](#)

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Question 45

Not yet answered

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 [Flag question](#)

A laboratory technologist performs Ziehl-Neelsen staining for the diagnosis of *Mycobacterium leprae*. The technologist performs all the stains correctly except adding the counter stain. What will the technologist likely observe during the examination of this staining?

- ☒ a. Red rod shaped bacteria with colorless background
- ☐ b. Green rod shaped bacteria with colorless background
- ☐ c. Green rod shaped bacteria with red background
- ☐ d. Red rod shaped bacteria with green background

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Question 44

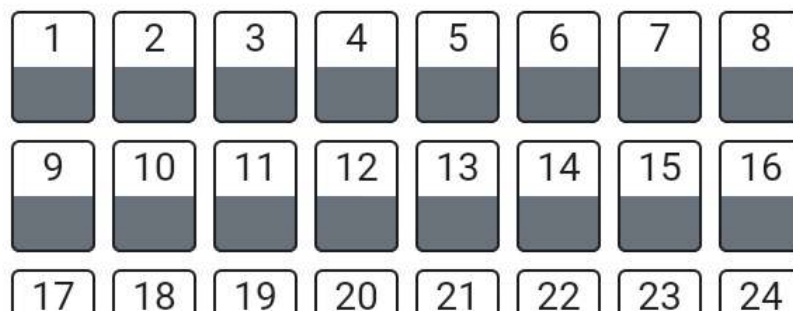
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 [Flag question](#)

A laboratory technologist performed routine urinalysis and obtained a positive nitrite test and a negative leukocyte esterase test. What pre-analytical error most likely causes this result?

- ☐ a. Vaginal discharge contamination
- ☐ b. Delay in specimen processing
- ☐ c. Container with preservative residue
- ☐ d. Inadequate mixing before examination

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 Flag question

A laboratory technologist performed appropriate dilution of whole blood to diluents for manual WBC count using Improved Neubauer chamber. Then, the technologist filled it into a clean dry counting chamber which was covered with its cover glass. The technologist is planning to start counting of the WBC in the one side of double cell Improved Neubauer ruled chamber. Where is the appropriate counting area for this?

- ☐ a. Four corner square of the ruled area
- ☐ b. The entire ruled area of the chamber
- ☐ c. Five center square of the ruled area
- ☐ d. All ruled area of the center square

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Question 42

Not yet answered

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 [Flag question](#)

A laboratory technologist is performing culture from throat swab collected from nine-year-old female patient. The laboratory technologist identified that there is *Streptococcus pyogenes* up on performing biochemical test. What will be their appearance up on gram staining?

- ☐ a. Gram negative cocci in chain
- ☐ b. Gram negative cocci in grapes
- ☐ c. Gram positive cocci in grapes
- ☐ d. Gram positive cocci in chain

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Question 41

Not yet answered

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[Flag question](#)

A laboratory technologist extracted and purified parasite DNA material from a given patient stool sample. Then the technologist started preparing a master mix solution to amplify the extracted DNA by the PCR method. Mg^{2+} ions are among the different components of the master mix solution, and so the technologist calculates the amount of Mg^{2+} ions needed to mix with the master mix solution. Which factor should be considered by the technologist to determine the Mg^{2+} ions concentration?

- ☐ a. The concentration of nucleotides, polymerase and primer used for PCR
- ☐ b. The atmospheric temperature during PCR reaction
- ☐ c. The bacteria strain from which template strand is extracted
- ☐ d. The type of electrophoresis used after PCR procedure

[Flag question](#)

A laboratory technologist received blood sample collected from a 16-year-old male patient for glucose analysis using spectrophotometer method. The instrument was under maintenance and the whole blood sample was left at room temperature for more than two hours prior to analysis. What would be the effect of delayed analysis on glucose concentration?

- ☐ a. Decrease glucose level due to lipogenesis at room temperature
- ☐ b. Decrease glucose level due to glycolysis at room temperature
- ☐ c. Decrease glucose level due to oxidation reaction at room temperature
- ☐ d. Decrease glucose level due to glycogenesis at room temperature

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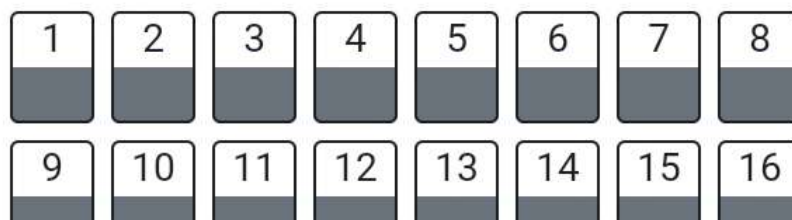
 [Flag question](#)

A laboratory technologist received a random urine specimen of a 24-year-old male patient with confirmed urinary schistosomiasis for routine urinalysis tests. The laboratory technologist analyzed and reported the urinalysis result to the physician. Which physical urine parameter was more helpful to diagnose the above case?

- ☐ a. Odor
- ☐ b. Specific gravity
- ☐ c. Volume
- ☒ d. Color

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Time left 2:36:31

Question 46

Not yet answered

Marked out of 1.00

 [Flag question](#)

A laboratory technologist planned to measure the ESR of a TB patient. To this effect, the technologist decided to collect 4 ml of blood in a test tube. What volume of citrate should be added in the test tube before collecting the blood?

- ☐ a. 36 ml
- ☐ b. 1 ml
- ☐ c. 16 ml
- ☐ d. 1.25 ml

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Time left 2:33:27

Question 51

Not yet answered

Marked out of 1.00

[Flag question](#)

A laboratory technologist received freshly voided morning urine from a 54-year-old male patient for routine urinalysis. The technologist examined the urine specimen physically and he reported the result as brown colored urine. However, after long standing, the color of the urine was changed to black color while the test was negative for urinary blood. Which urine parameter probably changed the urine color?

- ☐ a. Urinary bilirubin
- ☐ b. Urinary melanin
- ☐ c. Urinary hemoglobin
- ☐ d. Urinary glucose

Medical Laboratory Technology

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Time left 2:34:02

Question 50

Not yet answered

Marked out of 1.00

[Flag question](#)

A laboratory technologist received CSF specimen of a five-day-old female infant with confirmed xanthochromia due to immature liver development. The CSF specimen appeared significantly yellowish upon physical examination. What should the technologist determine next?

- ☐ a. Glucose and WBC count
- ☐ b. Bilirubin and RBC count
- ☐ c. Protein and WBC count
- ☐ d. Glucose and RBC count

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Time left 2:34:54

Question 49

Not yet answered

Marked out of 1.00

 [Flag question](#)

A laboratory technologist received clear pleural fluid specimen collected from inpatient ward. Then, the technologist determined the fluid-to-blood ratio for protein and lactic dehydrogenase. What is the most likely purpose of determining the ratio?

- ☐ a. To differentiate Cellular components
- ☐ b. To differentiate Bacterial infection from fungal infection
- ☐ c. To differentiate Transudate from exudate effusions
- ☐ d. To differentiate Viral infection from bacterial infection

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Time left 2:30:01

Question 56

Not yet answered

Marked out of 1.00

 [Flag question](#)

A laboratory technologist wants to perform manual white blood cell count using an improved Neubauer Counting Chamber for a 25-year-old pregnant mother who comes to a hospital for ANC follow-up. The technologist collects capillary blood and diluted with Turk's solution in 1:20 dilution. After filling the counting chamber and performing cell counts under 10 x objective light microscope, the technologist observes 23, 26, 22, and 21 WBCs in one side of the chamber; and 28, 24, 22, and 26 WBCs on the other side of the counting chamber. What is the total white blood cell count for the patient?

- ☐ a. 4.8×10^9 cells/L
- ☐ b. 9×10^9 cells/L
- ☐ c. 12×10^9 cells/L
- ☐ d. 3.8×10^9 cells/L

Medical Laboratory Science / Medical Laboratory Technology

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Time left 2:30:14

Question 55

Not yet answered

Marked out of 1.00

 [Flag question](#)

A laboratory technologist wanted to determine the specific gravity of urine specimen. He needs convenient method that removes the effect of glucose and urea on specific gravity measurement. What is the most appropriate method for this case?

- ☐ a. Harmonic oscillation densitometry
- ☐ b. Refractometry
- ☒ c. Urinometer
- ☐ d. Chemical reagent strip

[Clear my choice](#)

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Time left 2:30:53

Question 54

Not yet answered

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 [Flag question](#)

A laboratory technologist received urine specimen of a 40-year-old female patient for routine urinalysis tests. Nevertheless, he was busy by doing previously requested tests and he could not be able to process the sample in the recommended period. What is the best possible preservative to be used for the specimen?

- ☐ a. Freezing it at -20 degree centigrade
- ☐ b. Freezing it at -10 degree centigrade
- ☐ c. Refrigerating it at 2-8 degree centigrade
- ☐ d. Refrigerating it at 30 degree centigrade

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Time left 2:31:16

Question 53

Not yet answered

Marked out of 1.00

 [Flag question](#)

A laboratory technologist received urine specimen of a 13-year-old female patient with a history of urinary tract infection. The technologist analyzed the urine sample and found significant number of pyuria. What magnification is used to report this pyuria?

- ☐ a. Under low power field (10x)
- ☐ b. Under low power field (4x)
- ☐ c. Under high power oil immersion (100x)
- ☒ d. Under high power field (40x)

[Clear my choice](#)

Time left 2:32:04

Question 52

Not yet answered

Marked out of 1.00

 [Flag question](#)

A laboratory technologist received freshly voided urine of a 13-year-old male patient to check urine specific gravity. The technologist performed the test using both reagent strip and urinometer methods. Then, he found higher result in urinometer than the reagent strip. What could be the possible reason for the above result?

- ☒ a. Urinometer detects the density of ionic substance only
- ☐ b. Reagent strip detects the density of non-ionic substances only
- ☐ c. Urinometer detects the density of ionic and non-ionic substances
- ☐ d. Urinometer detects the density of non-ionic substance only

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Time left 2:27:43

Question 60

Not yet answered

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 [Flag question](#)

A laboratory technologist who is working at a district hospital laboratory collected blood specimen from a 48-year-old female patient for thyroid function tests. The tests measurement method was not available in the hospital and the specimen should be processed and transported to Regional Laboratory. What was the best specimen transportation method?

- ☐ a. Ice cold closed chain system
- ☐ b. Deep freezing system
- ☐ c. Liquid nitrogen system
- ☐ d. Room temperature system

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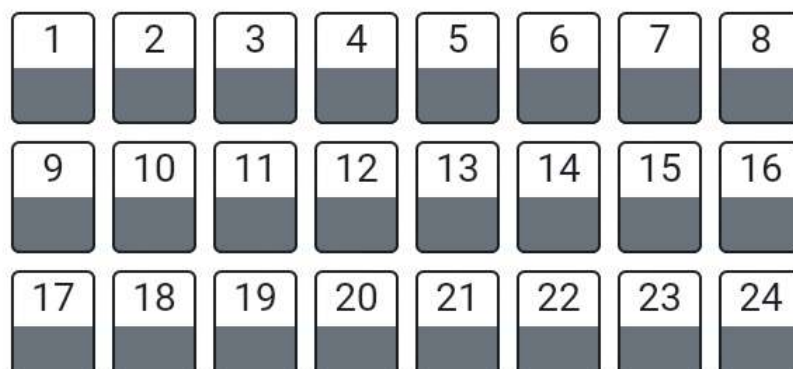
 [Flag question](#)

A laboratory technologist was examining dark field microscope for specimen collected from the genitalia lesion of sexual active male patient. The technologist observed thin, delicate spirochaete with 8–14 evenly sized coils. What is the most likely pathogen observed under the microscopy?

- ☐ a. *Leptospira interrogans*
- ☐ b. *Treponema pallidum*
- ☐ c. *Legionella pneumophila*
- ☐ d. *Borrelia burgdorferi*

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Time left 2:28:32

Question 58

Not yet answered

Marked out of 1.00

[Flag question](#)

A laboratory technologist was analyzing different clinical chemistry tests at Regional Laboratory using an open system clinical chemistry analyzer. The available reagents were stock out and other reagents were purchased for the analysis. Then, the instrument should be calibrated using appropriate solution. What would be the appropriate solution used to calibrate the instrument?

- ☐ a. Standard solutions
- ☐ b. Blank solutions
- ☐ c. Control solutions
- ☐ d. Known sample solutions

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Time left 2:29:28

Question 57

Not yet answered

Marked out of 1.00

[Flag question](#)

A laboratory technologist wants to perform manual white blood cell counting using an Improved Neubauer Counting Chamber and collects blood from a patient and dilute it with Turk's solution containing 2percent acetic acid. Then, the technologist transfers the diluted blood in to the counting chamber for counting. What does the 2percent acetic acid do in this procedure?

- ☐ a. It maintains the morphology of white blood cells
- ☐ b. It stains the nucleus of white blood cells
- ☐ c. It lyses non nucleated red blood cells
- ☐ d. It maintains the disc shape of red blood cells

Time left 2:24:03

Question 64

Not yet answered

Marked out of 1.00

 [Flag question](#)

A laboratory technologist working at urinalysis laboratory received urine specimen of a 34-year-old female patient for routine urinalysis test. Then, the technologist checked whether the specimen was properly labeled with the patient's identification number, date and time of collection, additional information such as patient's age, location, and physician's name immediately. What should the technologist do next before the actual examination process?

- ☒ a. Check the request paper
- ☐ b. Microscopic analysis
- ☐ c. Physical assessment
- ☐ d. Chemical analysis

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Medical Laboratory Technology

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Time left 2:25:36

Question 63

Not yet answered

Marked out of 1.00

[Flag question](#)

A laboratory technologist working at district laboratory received urine specimen of a 45-year-old male patient who was suspected for multiple myeloma. The technologist performed the analysis immediately. What urine parameter should be preferably measured for the patient?

- ☐ a. Nitrite
- ☐ b. Urine Bence Jones protein
- ☐ c. Urine creatinine
- ☐ d. Urine glucose



Not yet answered

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 [Flag question](#)

A laboratory technologist who is working in a hospital clinical chemistry laboratory bought AST reagents in two different bottles as reagent 1 and reagent 2. These stock solutions need to be converted to working solution by mixing 2 volumes of reagent 1 and 3 volumes of reagent 2 for one patient sample test which requires 1ml of working reagent. How much volume of reagent 1 and reagent 2 is needed to be mixed respectively?

- ☐ a. 0.6 ml ,0.4 ml
- ☐ b. 0.4 ml , 0.6 ml
- ☐ c. 0.2 ml , 0.8 ml
- ☐ d. 0.8ml , 0.2 ml

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Time left 2:26:13

Question 61

Not yet answered

Marked out of 1.00

 [Flag question](#)

A laboratory technologist who is working at a urinalysis laboratory received urine specimen of a 19-year-old female patient for routine urinalysis tests. Then, the technologist reported all normal and abnormal important urine microscopic parameters. Which objectives were used to report all necessary parameters?

- ☐ a. 10x and 20x
- ☒ b. 10x and 40x
- ☐ c. 40x and 100x
- ☐ d. 4x and 10x

[Clear my choice](#)

Medical Laboratory Science / Medical Laboratory Technology

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Time left 2:21:02

Question 67

Not yet answered

Marked out of 1.00

 [Flag question](#)

A medical laboratory scientist working at regional molecular biology laboratory forgot adding a primer to a master mix while running PCR amplification of a known segment of DNA. What would happen to the PCR result?

- ☐ a. Few non-specific amplifications
- ☐ b. Few amplification of segment of interest
- ☐ c. No problem with PCR amplification
- ☐ d. No amplification reaction

Time left 2:22:13

Question 66

Not yet answered

Marked out of 1.00

 [Flag question](#)

A laboratory technologist, who is working in a Clinical Chemistry Laboratory, was performing different clinical chemistry tests using automated clinical chemistry analyzer last week. In the second week, the technologist was running both normal and pathological controls for consecutive days, but their results were rejected. What should the technologist do next?

- ☐ a. Run reagent blank
- ☐ b. Run standard solution
- ☐ c. Run sample blank
- ☐ d. Run distilled water blank

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Time left 2:23:43

Question 65

Not yet answered

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 [Flag question](#)

A laboratory technologist, who has been working at Ethiopian Public Health Institute Laboratory, prepared a working reagent from the primary stain that would be used in acid fast staining procedure. What are the appropriate components of the primary stain to be used?

- ☐ a. Basic fuchsin, Phenol, Methanol, Distilled water
- ☐ b. Methylene blue, Potassium hydroxide, Ethanol, Distilled water
- ☐ c. Potassium iodide, Iodine, Methyl alcohol, Distilled water
- ☐ d. Crystal violet, Ammonium oxalate, Ethyl alcohol, Distilled water

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Time left 2:18:48

Question 69

Not yet answered

Marked out of 1.00

 [Flag question](#)

A medical laboratory technologist was involved in a research study to identify the most prevalent serogroup of *N.menegitidis* so that he collected specimen from some carriers and analyzed in the laboratory. The laboratory findings indicated that Serogroups Y and W135 are most widely distributed among the carriers. What is the most likely clinical condition caused by the serogroups?

- ☐ a. Meningococcemia
- ☐ b. Meningococcal pharyngitis
- ☐ c. Meningococcal meningitis
- ☐ d. Meningococcal pneumonia

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Time left 2:19:34

Question 68

Not yet answered

Marked out of 1.00

 [Flag question](#)

A medical laboratory technologist collected a random blood specimen to analyze blood glucose level. He knows that clinically significant glucose changes occur if serum remains in prolonged contact with blood cells without preservatives and should use appropriate preservative to avoid such condition. What is the more appropriate preservative for this case?

- ☐ a. Sodium fluoride
- ☐ b. Lithium heparin
- ☐ c. Potassium oxalate
- ☐ d. Sodium heparin

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Time left 2:16:52

Question 71

Not yet answered

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 [Flag question](#)

A molecular biologist was running a PCR that involved use of several pairs of primers to amplify more than one DNA fragment in the same reaction to diagnose for enteric pathogens in stool samples. What type of PCR was most likely used?

- ☐ a. Real time PCR
- ☐ b. Multiplex PCR
- ☐ c. Nested PCR
- ☐ d. Reverse transcriptase-PCR

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Time left 2:18:20

Question 70

Not yet answered

Marked out of 1.00

[Flag question](#)

A medical parasitologist wants to screen under-five children for hemoparasites in a rural district. He prepares both thick and thin blood smears from skin puncture. Then, he fixes the thin smears in batch with absolute methanol before transporting it to the Medical Parasitology Laboratory for microscopic examination. What is the role of fixation in the procedure?

- ☐ a. It keeps the blood films free from dust
- ☐ b. It preserves the morphology of cells
- ☐ c. It prevents the blood film from scratches
- ☐ d. It lyses the unwanted cells in the film

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Time left 2:13:18

Question 72

Not yet answered

Marked out of 1.00

[Flag question](#)

A molecular biologist who is working at a regional laboratory was running PCR for the diagnosis of infectious diseases and wanted to monitor PCR product accumulations while the PCR is still running. Which type of PCR can satisfy the need of this molecular biologist?

- ☐ a. Nested PCR
- ☒ b. Real-time PCR
- ☐ c. Qualitative PCR
- ☐ d. Multiplex PCR

[Clear my choice](#)



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Time left 2:13:39

Question 71

Not yet answered

Marked out of 1.00

 [Flag question](#)

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- ☐ b. Multiplex PCR
- ☐ c. Nested PCR
- ☒ d. Reverse transcriptase-PCR

[Clear my choice](#)

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A medical parasitologist wants to screen under-five children for hemoparasites in a rural district. He prepares both thick and thin blood smears from skin puncture. Then, he fixes the thin smears in batch with absolute methanol before transporting it to the Medical Parasitology Laboratory for microscopic examination. What is the role of fixation in the procedure?

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- ☒ b. It preserves the morphology of cells
- ☐ c. It prevents the blood film from scratches
- ☐ d. It lyses the unwanted cells in the film

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Time left 2:12:18

Question 74

Not yet answered

Marked out of 1.00

 [Flag question](#)

A newly graduated laboratory technologist received a blood specimen from a patient who came from an area having haemorrhagic fever outbreak. Electron micrograph show long thread like appearance. When he requested to do cell culture, the senior recommends avoiding doing cell culture as the isolate could be very dangerous. What is the most likely probable causative agent?

- ☐ a. Ebola
- ☐ b. Measles
- ☐ c. Hanta virus
- ☐ d. Influenza

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Time left 2:12:57

Question 73

Not yet answered

Marked out of 1.00

[Flag question](#)

A molecular biologist who is working at diagnostic molecular laboratory was planning to observe the image of DNA bands on a gel electrophoresis from the result of PCR amplicons of patients samples with the help of UV light. He forgot to add Ethidium Bromide to a DNA solution. What is the most probable effect of this mistake?

- ☐ a. The ladder bands will be visible
- ☐ b. No bands will be seen irrespective of patients infection status
- ☐ c. Negative samples will have no bands on the gel
- ☐ d. Positive sample will show bands on the gel

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Time left 2:09:12

Question 77

Not yet answered

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 [Flag question](#)

A physician collected bone marrow biopsy from the iliac crest of a 50-year-old patient diagnosed with CML. As soon as the bone marrow sample was collected, the physician prepared smears and sent them to the laboratory. The laboratory technologist stained the smear and examined it microscopically and found bone marrow cellularity of 80percent. What is the interpretation of the result?

- ☐ a. The marrow is hypocellular
- ☐ b. The marrow is normocellular
- ☐ c. The marrow is hypercellular
- ☐ d. The marrow is filled with excess fat cells

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Time left 2:09:40

Question 76

Not yet answered

Marked out of 1.00

 [Flag question](#)

A Phlebotomist wants to draw a blood sample from a vein of a child admitted to a pediatric ward; and to ship it to the hospital laboratory for blood chemistry investigation. What is the ideal blood collecting system?

- ☐ a. Micro-hematocrit system
- ☐ b. Syringe system
- ☐ c. Evacuated tube system
- ☐ d. Butterfly system

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Time left 2:11:13

Question 75

Not yet answered

Marked out of 1.00

 [Flag question](#)

A patient came to a hematology department with a request paper for white blood cell count. The routine test performed in the laboratory was manual cell counting using Improved Neubauer chamber. The technologist wants to select the diluents used for WBC count. What is the most appropriate diluent for this test?

- ☐ a. Tape water
- ☐ b. Acetic acid
- ☐ c. Normal saline
- ☐ d. Distilled water

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Time left 2:07:39

Question 79

Not yet answered

Marked out of 1.00

 [Flag question](#)

A physician wanted to know the complete blood cell count of a patient with a history of chronic smoking. The patient came to the laboratory and gave venous blood. The laboratory technologist analyzed the complete blood count using a five part automated haematology analyzer and reported the result to the physician. Which haematological parameters are likely to be elevated in the report?

- ☐ a. Total white cell count
- ☐ b. Absolute Lymphocyte count
- ☒ c. Platelet count, PDW and Pct
- ☐ d. Hgb, Hct, MCV

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Time left 2:08:30

Question 78

Not yet answered

Marked out of 1.00

 [Flag question](#)

A physician diagnosed a 60-year-old patient with blurred vision and cervical stiffens. The physician collected CSF specimen of the patient in a single test tube because of the shortage of sterile test tubes. What was most likely affected by the above procedure?

- ☐ a. Cryptococcal antigen test
- ☐ b. Cell count
- ☐ c. Glucose determination
- ☐ d. Protein determination

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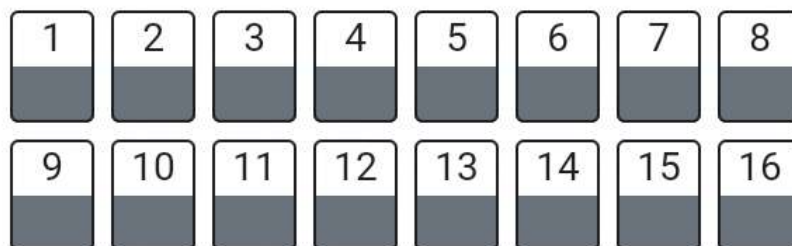


A sputum sample collected from known pulmonary tuberculosis patient is sent for an external quality assessment to a health center laboratory. The laboratory technologist at the health center performs the acid fast staining correctly. What is the expected result in this microscopy examination of AFS?

- ☐ a. Blue rod bacteria with colorless background
- ☐ b. Blue rod bacteria with red background
- ☐ c. Red rod bacteria with colorless background
- ☐ d. Red rod bacteria with blue background

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Time left 2:06:30

Question 81

Not yet answered

Marked out of 1.00

 [Flag question](#)

A six-month-old baby girl came to a laboratory for blood film examination. A laboratory technician has received the request paper and planned to collect blood sample from the patient. What is the most appropriate site of blood collection for this patient?

- ☐ a. Palmer surface of the ring finger of the baby
- ☐ b. Median cubital vein on the forearm of the baby
- ☐ c. Plantar surface of the heel of the baby
- ☐ d. Cephalic vein on the forearm of the baby

Not yet answered

Marked out of 1.00

 [Flag question](#)

A researcher wanted to assess urinary bilirubin level among study subjects with severe *P.falciparum* malaria. He analyzed urine bilirubin using both the reagent strip (diazotization) and test tube (Fouchet's test) methods. Then, the reagent strip test was positive for all participants while the test tube test was negative. What could be the possible reason for the above results?

- ☒ a. Reagent strip test is not appropriate for urinary bilirubin assessment
- ☐ b. Both tests are not appropriate tests for urinary bilirubin assessment
- ☐ c. Fouchet's reagent may lack ferric chloride (iron III chloride)
- ☐ d. Bilirubin may not be found in the urine

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Time left 2:03:51

Question 85

Not yet answered

Marked out of 1.00

 [Flag question](#)

A watery diarrhea incidence was occurred in a given town of the country. The laboratory personnel participated in route cause analysis of the diarrhea. During investigation, *Vibrio cholerae* was identified as etiological agent of the outbreak in the town. What is the most likely mode of transmission of the identified pathogen?

- ☐ a. Fecal-oral
- ☐ b. Vector
- ☐ c. Fomites
- ☐ d. Aerosol

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Time left 2:04:12

Question 84

Not yet answered

Marked out of 1.00

 [Flag question](#)

A two-month-old baby boy developed congenital defects like congenital heart defect, hearing loss and cataract (vision problem). He also manifested with maculopapular rash which begins on the face and moves down the body. Small size, envelope surrounded icosahedral nucleocapsid with positive sense single stranded RNA genome was isolated using molecular and microgram analysis. What is the most likely disease the child has developed?

- ☐ a. Corona
- ☐ b. Rubeola
- ☐ c. Mumps
- ☐ d. Rubella

Time left 2:04:28

Question 83

Not yet answered

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 [Flag question](#)

A three-year-old child developed serous rhinorrhea, sneezing, anorexia, and low-grade fever, progressing to the repetitive coughs followed by inspiratory whoops which end up with vomiting and exhaustion. There was a marked lymphocytosis in complete blood count analysis. Nasopharyngeal swab was taken and inoculated in a culture media enriched with charcoal, starch, blood, and albumin; the result revealed extremely small gram-negative coccobacilli. What is the most likely etiologic agent for the patient condition?

- ☐ a. *Bordetella pertussis*
- ☐ b. *Chlamydophila pneumoniae*
- ☐ c. *Mycoplasma pneumonia*
- ☐ d. *Legionella pneumophila*

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Time left 2:00:44

Question 89

Not yet answered

Marked out of 1.00

 [Flag question](#)

Before using for PCR analysis, both DNA and RNA purity assessment is required. A laboratory technologist wanted to do purity assessment on a preparation of DNA. What value dose he use as an indicator of the relative purity?

- ☐ a. Absorbance at both 280
- ☐ b. The ratio of absorbance at 260 and 280 nm
- ☐ c. Absorbance at both 260
- ☒ d. Absorbance at both 270

[Clear my choice](#)

Time left 2:01:28

Question 88

Not yet answered

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 [Flag question](#)

A term, male, 3300 grams weighing infant was delivered spontaneously. Physical examinations of the infant were normal during the first week of life; however, the child started feeding irregularly during the second week. On day 13, the baby was admitted to the hospital with generalized seizures. A small amount of cloudy cerebrospinal fluid was collected by lumbar puncture, and a group B Streptococcus was isolated from culture. What was the most likely disease the infant developed?

- ☐ a. Subacute endocarditis
- ☐ b. Late onset neonatal sepsis
- ☒ c. Rheumatic fever
- ☐ d. Acute glomerulonephritis

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Question 87

Not yet answered

Marked out of 1.00

 [Flag question](#)

An 80-year-old diabetic patient monitors his blood glucose level regularly. During his follow up, urine chemical analysis has been performed every month. What type of urine specimen is most appropriate?

- ☐ a. Catheterized urine specimen
- ☐ b. Midstream urine specimen
- ☐ c. 24-hour urine specimen
- ☒ d. Fasting urine specimen

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Question 86

Not yet answered

Marked out of 1.00

 [Flag question](#)

About 150 bags of blood were collected through community mobilization and transported to the blood bank. The laboratory technologist working in a blood bank conducted blood grouping and planned to screen the collected blood for HIV, Hepatitis B and C Virus. What is the most appropriate screening test to be used?

- ☐ a. Latex agglutination (LA) test
- ☐ b. Indirect fluorescent antibody (IFA) test
- ☐ c. Radio immune assay (RIA) test
- ☐ d. Enzyme linked immune sorbent assay (ELSA) test

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Question 93

Not yet answered

Marked out of 1.00

 [Flag question](#)

In RBC morphological study, a laboratory technologist performed the Wright's stain after properly prepared thin blood films. Then, the technologist examined under oil immersion objective. The technologist found RBC size which is equal to the nucleus of small lymphocyte and its shape looks like biconcave disc. What is the most likely interpretation for this finding?

- ☐ a. Decrease in RBC size, but Normal shape
- ☒ b. Normal RBC size and shape
- ☐ c. Normal RBC size, but abnormal shape
- ☐ d. Increase in RBC size, but normal shape

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Time left 1:58:35

Question 92

Not yet answered

Marked out of 1.00

 [Flag question](#)

If a chemistry analyst mixes a p-Nitrophenylphosphate, a color less compound, used as a reagent with a patient sample, a yellow p-nitrophenol product that absorbs light at 405nm is produced due to hydrolysis of the reagent in patient's sample; and the amount of light absorbed by this product is directly proportional to the activity of the enzyme to be measured. Which enzyme is measured based on the above test principle?

- ☐ a. Acid phosphatase(ACP)
- ☐ b. Gamma glutamyltransferase (GGT)
- ☐ c. 5-Nucleotidase (5NT)
- ☒ d. Alkaline phosphatase (ALP)

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Question 91

Not yet answered

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During solid culture medium preparation, a laboratory technologist plans to kill contaminant microorganisms in the culture media. The manufacturer instructs the medium to be sterilized by steam under pressure. What method of sterilization is appropriate for the medium?

- ☐ a. Boiling
- ☐ b. Steaming
- ☐ c. Filtration
- ☒ d. Autoclaving

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Question 90

Not yet answered

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 [Flag question](#)

CSF specimen of a 30-year-old patient was delivered to the hematology laboratory. The hematologist observed that the specimen was clear and colorless. What will be done in the next step by the hematology laboratory?

- ☐ a. Store the specimen
- ☐ b. Request other new specimen
- ☒ c. Perform cell count
- ☐ d. Report the result as normal

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Question 97

Not yet answered

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Proper collection and handling of samples for K⁺ analysis is extremely important to address quality laboratory result for K⁺ electrolyte measurement as there are many artifactual factors that falsely elevate its serum level in the patient's sample. These factors can cause falsely elevated serum potassium level or artifactual hyperkalemia. Which factor is commonly the cause for the above condition?

- ☒ a. Prolonged tourniquet
- ☐ b. Thrombocytosis
- ☐ c. Frozen samples
- ☐ d. Hemolysis

[Clear my choice](#)

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 [Flag question](#)

Previously, in a hospital, viral infection diagnosis was being made based on clinical manifestation. Currently, the hospital was provided with different serological test kits for demonstration of specific viral antibodies in the patient's sera. What is the most likely limitation of using provided serological test kits?

- ☐ a. Used for diagnosis of infections caused by viruses difficult to cultivate
- ☐ b. Serological tests may be associated with false-positive or false-negative reaction
- ☐ c. During first weeks of primary infection viral antibodies in the serum were detected
- ☐ d. Detection of virus-specific antibody in serum indicates a recent primary infection

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Question 95

Not yet answered

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Outbreak occurs at a day care centre in which most of the infants initially develop low grade fever followed by persistent vomiting and diarrhea. The infants appeared dehydrated and required hospitalization. The surveillance team confirmed it is caused by rotavirus. What could be the most probable source of the infection?

- ☐ a. Contaminated sharp
- ☒ b. Infected animal
- ☐ c. Contaminated air
- ☐ d. Common water source

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[Flag question](#)

Newly prepared acid fast staining reagents (Carbol fuchsin and methylene blue) have been distributed by a regional laboratory to different laboratories of hospitals and health centers. Each facility that received the reagents is told to check the final quality phase of reagents prior to applying them for patient examination. What is the most appropriate method for checking the quality of reagents in each facility?

- ☐ a. perform using appropriate control samples
- ☐ b. Correct calculation of the ingredients
- ☒ c. Inspect the physical appearance of the reagents
- ☐ d. Proper selection of measuring material

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Question 100

Not yet answered

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While conducting a clinical chemistry practical session, a laboratory assistant mixes serum with alkaline picrate and let the students to measure the rate of change of absorbance in semi-automated spectrophotometer; and the rate of change of absorbance becomes 1.4mg/dl after measurement. Which renal marker is more likely measured by this test principle?

- ☐ a. Uric acid
- ☐ b. Creatinine
- ☐ c. Cystatin C
- ☐ d. Urea

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Question 99

Not yet answered

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There is a project on plasma proteins level assessment among youngsters, adults and elders for research purpose. A laboratory technologist working in a research laboratory performs a total protein determination by Kjeldahl method. Which component of protein is determined by Kjeldahl method?

- ☐ a. Hydrogen
- ☒ b. Nitrogen
- ☐ c. Sulfur
- ☐ d. Carbon

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Question 98

Not yet answered

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The Clinical Chemistry course Team received a newly purchased Chemistry Analyzer Machine, Spectrophotometer, for its students to practice on it after it is fully installed. Before using the analyzer for chemistry tests, the photometric and wavelength accuracy; and photometric linearity of the analyzer must be checked. Accordingly, when the Clinical Chemistry analyst checks these parameters, he identified problems with photometric accuracy of the machine. What is the most appropriate mechanism to fix this problem?

- ☐ a. Replacing the excitor lamp
- ☐ b. Cleaning a photocell window
- ☐ c. Testing for excess stray light
- ☐ d. Realigning excitor lamp